# MUHAMMED SAAHIL

Game Developer | Unity & Unreal Engine 5 | Technical Art

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## PROFESSIONAL SUMMARY

Game Developer currently in the 3rd year of a degree in Game Designing and Development, with hands-on experience building gameplay systems and prototypes in Unity and Unreal Engine 5. Experienced across gameplay programming, player controllers, physics, camera systems, multiplayer concepts, materials, shaders, VFX, level prototyping, and rapid game-jam development. Separately from my game projects, I have practical experience in 3D modeling, UV mapping, texturing, and environment/prop creation using Blender, Maya, and Substance Painter. Also developed LevelLens, a web-based player-behavior analytics tool processing Apache Parquet telemetry, combining a developer's implementation perspective with a level designer's analytical perspective. Seeking opportunities in gameplay programming, technical implementation, prototyping, and 3D/game development.

## SKILLS

**Game Engines:** Unity, Unreal Engine 5
**Programming & Tools:** C#, Blueprint scripting, Git/version control
**Gameplay Systems:** Player controllers, combat systems, physics-based gameplay, camera systems (Cinemachine), interaction systems, multiplayer/co-op concepts, level prototyping
**Technical Art / Shaders:** Shader Graph, URP, Unreal Materials & Material Instances, emissive effects, Niagara/VFX, Chaos physics & geometry fracturing, PBR materials
**3D Art (standalone work):** Blender, Maya, Substance Painter, prop/environment modeling, texturing
**Game Analytics:** Player telemetry, heatmaps, movement visualization, route analysis, death-location analysis, minimap visualization
**Design:** Game design, level design concepts, puzzle & platformer design, game loops, rapid prototyping

## FEATURED PROJECT

**Supermarket Game — Game Developer**
*Tools/Stack: Unity, Gameplay Programming, Environment/Level Construction*

- Leading current/most recent project: a supermarket-based game marking a transition from small game-jam prototypes to a larger, production-style build.
- Applying gameplay programming, environment development, level construction, and interaction systems to build out core store mechanics and player experience.
- Positioned as one of the strongest pieces in the portfolio as systems and presentation are polished.

## GAME DEVELOPMENT PROJECTS

**GRID-LOCKED: The 99-Tile Blitz — Game Developer / Designer / Technical Artist**
*Tools/Stack: Unreal Engine 5, Blueprints, Materials, Chaos Physics*

- Designed a 2.5D isometric competitive multiplayer concept centered on a 99-tile territory-control arena.
- Built a material/shader system that dynamically updates tile ownership visuals (color transitions, glowing and blinking lines) while preserving the base tile design.
- Prototyped destructible geometry using Unreal Engine 5's Chaos physics system, including fracture simulation testing, alongside Blueprint-driven character movement and gameplay logic.

**Stay With Me — Game Developer**
*Tools/Stack: Unity, Split-screen Cameras, Shaders*

- Developed a two-player split-screen co-op/puzzle game with contrasting light/dark visual themes for each player.
- Implemented a shader-based, rotatable screen divider that turns the split-screen boundary itself into a gameplay mechanic.

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**3D Climbing Platformer (Brackeys Game Jam) — Game Developer**
*Tools/Stack: Unity*
- Developed a vertical 3D platforming prototype focused on climbing and traversal, where the goal is to keep climbing upward through the level.
- Implemented custom character movement, climbing interactions, physics-based falling, and vertically structured level progression with increasing difficulty.

**Big Hero 6–Inspired 3D Action Game — Game Developer / Animator**
*Tools/Stack: Unity*
- Built a 3D city environment and gameplay around a Baymax-robot-takeover premise, contributing across Unity development and game design.
- Documented the project through academic/project reporting.

**MaskOut — Game Developer**
*Tools/Stack: Unity, 2D Puzzle Platformer*
- Designed a mask-based traversal mechanic (Player → Mask → Environment) driving platform puzzles, path creation, and navigation challenges in a game-jam setting.

**Boon & Bane — Game Developer**
*Tools/Stack: Card/Tapping Mechanics*
- Translated a game-jam theme (boon/bane) into a card-and-tapping gameplay system where player actions trigger beneficial or detrimental consequences.

**Mobile Gyro Obstacle Endless Runner — Game Developer**
*Tools/Stack: Unity, Mobile, Gyroscope Input*
- Built a mobile endless runner using gyroscope-based tilt controls in place of touch input, including obstacle generation, collision, and progression difficulty.

**2D Puzzle Platformer (Prototype) — Game/Level Developer**
*Tools/Stack: Unity, Mobile*
- Designed and prototyped a 2D puzzle-platformer for a business/product concept, covering movement, puzzle mechanics, platforming, and mobile controls.

**Reversal Paradox — Game Developer / Level Designer**
*Tools/Stack: Experimental Prototype*
- Prototyped an experimental mechanic reversing standard player expectations and cause-and-effect relationships in rapid game-jam development.

## TOOL DEVELOPMENT

**LevelLens — Player Behavior Analysis Tool — Developer / Tool Developer**
*Tools/Stack: Next.js/React, FastAPI, Apache Parquet*
- Built a full-stack analytics web application that converts raw player telemetry (position, timestamp, match, map, death, and movement events) into visual insights for level designers.
- Designed a FastAPI backend with endpoints for loading Parquet files, decoding events, detecting maps/dates/matches, and extracting player positions.
- Developed a coordinate-mapping system converting in-game world coordinates (X/Y/Z) into minimap pixel coordinates, enabling accurate overlay of player movement on level maps.
- Implemented a grid-based heatmap pipeline (position → coordinate mapping → grid bucketing → frequency calculation), tuning grid resolution (~40 cells) to produce localized, actionable hotspots instead of broad blobs.
- Surfaced frequent locations, death locations, common routes, and movement paths across multiple maps (e.g., Ambrose Valley, Grand Rift, Lockdown), architected to scale to additional datasets without manual per-map configuration.

## EDUCATION

**B.Sc./B.A. in Game Designing and Development** (in progress — 3rd year)
ICAT Design & Media College | Expected graduation: April – July 2027

**Higher Secondary (12th)** — 80%
**Secondary School (10th), Kerala Board** — 89%

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